CSE 455/555
Introduction to
Pattern Recognition

Deepfake Face Detection via β –Variational Auto Encoder Latent Pattern Recognition

Jayachandra Galda | Gayathri Adulla| Saakketh Gouti
Instructor: Dr. Ifeoma Nwogu

Introduction

The rapid evolution of generative artificial intelligence has enabled the creation of highly realistic synthetic facial imagery, commonly referred to as *deepfakes*. While these technologies offer significant creative and commercial potential, they also introduce serious risks in digital forensics, biometric security, identity fraud, and large-scale misinformation. Accurately distinguishing real human faces from AI-generated images remains a challenging pattern recognition problem due to the visual realism and diversity of modern generative models.
In this work, we propose a **hybrid deepfake detection framework that integrates unsupervised feature learning using a β-Variational Autoencoder (β-VAE) with supervised classification in the learned latent space**. Our approach leverages the representational power of unsupervised learning while achieving high discriminative performance through lightweight supervised modeling.

Data Collection

Real Face Dataset

Authentic facial images were sourced from the Flickr-Faces-HQ (FFHQ) dataset, a large-scale, high-quality collection of real human face images. The dataset provides substantial diversity across **age, gender, ethnicity, pose, illumination, facial expressions, and background complexity**, making it ideal for learning robust real-face distributions.

Synthetic Fake Face Generation

Fake images were generated using a diffusion-based image synthesis model, the SG161222/Realistic_Vision_V5.1_noVAE Stable Diffusion model. A carefully engineered hyper-realistic facial portrait prompt was used to synthesize realistic human faces. Image generation was performed at **512×512 resolution** using **28 inference steps** and a **guidance scale of 3.0**, producing high-fidelity synthetic samples with realistic skin textures and lighting cues.

Final Dataset Summary

Real Images: 5,012
Fake Images: 1,800
Total Samples: 6,812
All images were resized to **64×64**, normalized, and converted into tensor format prior to training and evaluation.

Methods

β-VAE Latent Feature Learning

A **β-Variational Autoencoder (β-VAE)** is employed to learn compact and structured representations of facial images. The architecture consists of:
Encoder: Three convolutional layers for hierarchical spatial feature extraction
Latent Space: 128-dimensional continuous embedding
Decoder: Symmetric transposed convolution network for image reconstruction
The optimization objective is defined as:
L = MSE + β · KL || p,
with **β = 4**, enforcing strong regularization and disentanglement in the latent space.

Latent Space Classification

After training the β-VAE in an unsupervised manner, the learned **128-dimensional latent vectors** are extracted and used as feature representations for supervised classification. A **Logistic Regression classifier** is trained under the following settings:
Train/Test Split: 80/20 (stratified)
Feature Standardization: Standard Scaler
Target Labels: Real (0), Fake (1)
This approach converts the unsupervised latent space into a highly discriminative representation.

Experiments

Three experimental stages were conducted:

1. Unsupervised Representation Learning:
The β-VAE was trained for 50 epochs with early stopping on the combined dataset. Stable convergence was achieved without posterior collapse.

2. KL-Divergence Based Anomaly Detection:
A threshold based on the **95th percentile of real-image KL divergence** was applied for unsupervised detection. While this method achieved strong real-image detection, it failed to detect most fake samples due to distribution overlap.

3. Supervised Latent Classification:
Supervised learning on the latent space achieved **near-perfect discrimination** between real and fake classes.
Additionally, **t-SNE visualization** was used to project the latent embeddings into two dimensions, revealing clear nonlinear separability between the two classes.

Figure 1. a)Real image *b) AI generated image*

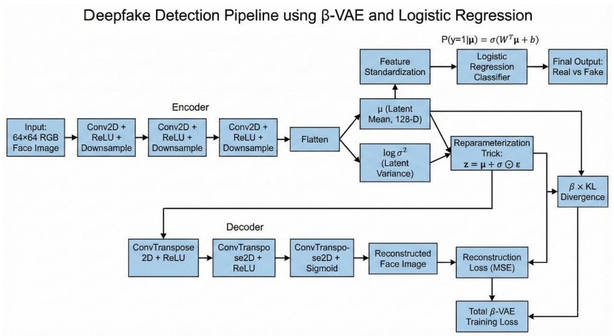

Figure 2. High level architecture of the pipeline

Evaluation

The final supervised model achieved the following performance:

- **Overall Accuracy: 98%**
- **Fake Precision: 95%**
- **Fake Recall: 97%**
- **ROC-AUC: 0.997**

The confusion matrix demonstrates minimal misclassification, and the ROC curve confirms excellent class separability within the learned latent space.

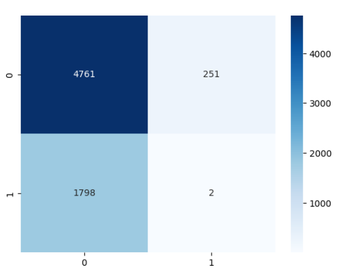

Figure 3. Confusion matrix

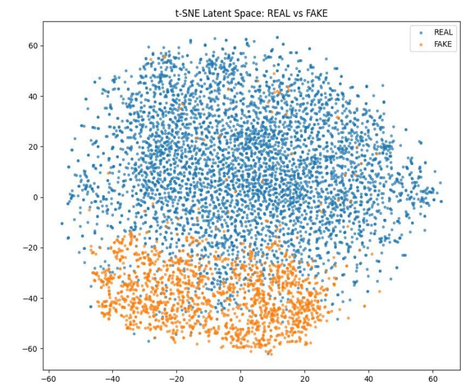

Figure 5. t-SNE visualization of 128-dimensional latent space

Conclusion

This study demonstrates that **combining unsupervised β-VAE representation learning with supervised latent space classification provides a highly accurate and robust deepfake detection framework**.
While unsupervised KL-divergence thresholding alone is insufficient when trained on mixed data, supervised learning on structured latent embeddings successfully overcomes this limitation and achieves near-optimal classification performance.

Future work will extend this framework toward:
·Video-based deepfake detection
·Temporal and attention-based modeling
·Real-time deployment in digital forensic and biometric security systems.

References

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University at Buffalo
Department of Computer Science and Engineering
School of Engineering and Applied Sciences